Applied SVAR Identification

Day 5: Identification IV — Instruments

Jamie L. Cross

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Course Roadmap

Day	Topic
1	VAR Foundations: Motivation, Estimation, Specification, Identification & Tools
2	Identification I: Short-Run Restrictions
3	Identification II: Long-run Restrictions
4	Identification III: Sign, Magnitude, and Narrative Restrictions
5	Identification IV: Instruments

Learning Objectives

By the end of today, you should be able to:

1. Explain how external instruments solve the simultaneity problem in SVARs.
2. Understand why instrument *relevance* and *exogeneity* are both required for valid identification.
3. Describe how high-frequency identification constructs valid instruments from asset price movements around scheduled announcements, and trace the approach from Kuttner (2001) to modern proxy SVARs.
4. Implement proxy SVAR identification in MATLAB using the ACB Toolbox.
5. Replicate the key results of Gertler and Karadi (2015, AEJ:Macro).
6. Describe the exogenous variable (VARX) approach and its equivalence to proxy SVARs.

Part 1

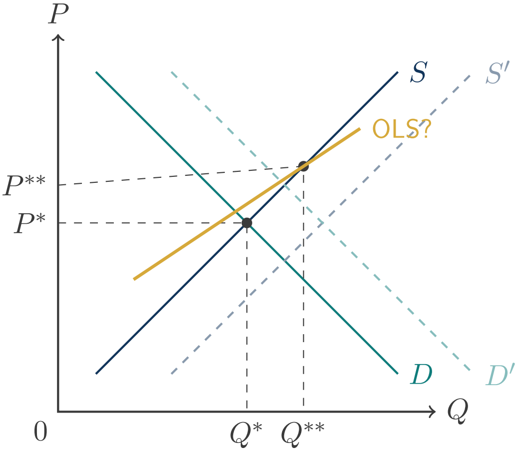
IV Intuition

Solving simultaneity with external variation

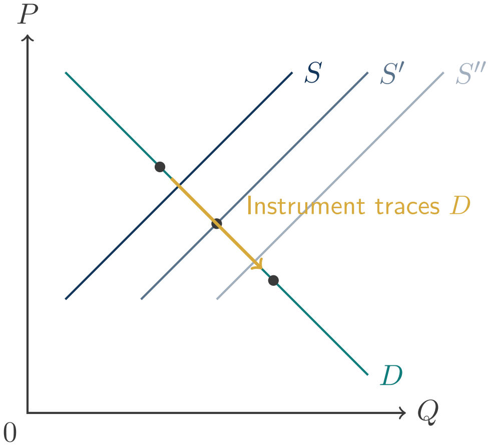
Background: Identification so far

- **Short-run restrictions**: Recursive ordering (Cholesky) imposes zero contemporaneous effects — arbitrary and sensitive to variable order
- **Long-run restrictions**: Impose zero long-run multipliers — credible only for specific structural questions (e.g. permanent demand shocks)
- **Sign restrictions**: Use theory to restrict the *direction* of responses — set identified, wide uncertainty bands
- **Recurring problem for monetary policy**: Recursive identification produces the **price puzzle** — prices *rise* after a contractionary shock. A sign of misidentification.
- **Another idea**: Borrow from microeconometrics. Use a variable *external* to the VAR that is correlated with the shock of interest but uncorrelated with all other shocks (Stock and Watson, 2012; Mertens and Ravn, 2013)

The simultaneity problem: a supply and demand example



Both S and D shift → OLS identifies **neither** curve



Instrument shifts S only → equilibria trace out **demand** D

The instrumental variable (IV) solution: formal requirements

- We trace the demand curve using a variable that shifts *only* supply — called an **exogenous instrument for the supply shock**. Two conditions must hold:
  1. **Relevance**: z_t is correlated with the supply shock ε_t^S:
  $$\mathbb{E}[z_t \varepsilon_t^S] = c \neq 0$$
  2. **Exogeneity**: z_t is uncorrelated with the demand shock ε_t^D:
  $$\mathbb{E}[z_t \varepsilon_t^D] = 0$$
- Test relevance using 'first-stage' F-statistic (generalizes to multiple instruments): $F > 10$ is considered strong evidence (Stock and Yogo (2002); Stock, Wright and Yogo (2002)).
- Exogeneity is **not directly testable** — ε_t^D is unobserved, so $\mathbb{E}[z_t \varepsilon_t^D] = 0$ cannot be verified directly.
- **Classic example**: *Weather* shifts agricultural supply (relevance) without affecting consumer preferences (exogeneity) — see Abbring, Chernozhukov, Fernández-Val (2025) attribute this IV strategy (among other things) to Wright (1928).

From markets to SVARs: the proxy SVAR logic

Back to the demand and supply identification problem:

$$\begin{pmatrix} e_{Pt} \\ e_{Qt} \end{pmatrix} = \underbrace{\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}}_{B_0^{-1}} \begin{pmatrix} \varepsilon_t^S \\ \varepsilon_t^D \end{pmatrix}$$

Instrument z_t shifts only supply: $\mathbb{E}[z_t \varepsilon_t^S] = c \neq 0$ and $\mathbb{E}[z_t \varepsilon_t^D] = 0$. So:

$$\mathbb{E}[e_{Pt} \cdot z_t] = b_{11}\, c, \qquad \mathbb{E}[e_{Qt} \cdot z_t] = b_{21}\, c$$

Dividing out c:

$$\underbrace{\frac{b_{21}}{b_{11}}}_{\text{identify supply shock}} = \frac{\mathbb{E}[e_{Qt} \cdot z_t]}{\mathbb{E}[e_{Pt} \cdot z_t]} = \left.\frac{\Delta Q}{\Delta P}\right|_{\text{supply shift}} = \underbrace{\textbf{demand slope}}_{\text{trace demand curve}}$$

Key Idea

In an n-variable SVAR, the same argument identifies **all** ratios b_{j1}/b_{11}: the full first column of B_0^{-1}, with dynamic responses then following from the estimated VAR.

Two-stage implementation

1. **Estimate the reduced-form VAR**, obtaining residuals $\hat{e}_t = (\hat{e}_{Pt},\, \hat{e}_{Qt})'$

2. **First stage**: Regress the *price* residual on the instrument

$$\hat{e}_{Pt} = \alpha + \beta z_t + \xi_t$$

   Save fitted values $\hat{e}^S_{Pt} = \hat{\alpha} + \hat{\beta} z_t$. Test instrument strength: F-stat > 10.

3. **Second stage**: Regress the *quantity* residual on $\hat{e}^S_{Pt}$

$$\hat{e}_{Qt} = \gamma\, \hat{e}^S_{Pt} + \zeta_t$$

   Coefficient $\hat{\gamma}$ identifies b_{21}/b_{11} — the **demand slope**

4. **Normalize** to an economically meaningful shock size
5. **Compute IRFs** given this vector

General implementation: n-variables

1. **Estimate the reduced-form VAR** and obtain residuals $\hat{e}_t$
2. **First stage**: Regress $\hat{e}_{1t}$ on z_t; save $\hat{e}^S_{1t}$. Test F-stat > 10.
3. **Second stage**: Run $n-1$ regressions:

$$\hat{e}_{jt} = \gamma_j \, \hat{e}^S_{1t} + \zeta_{jt}, \qquad j = 2, \ldots, n$$

   Each $\hat{\gamma}_j$ estimates the ratio b_{j1}/b_{11}

4. **Recover scale**: normalize b_{11} (e.g. set to 1) to obtain the full first column $\hat{b}_1 = b_{11}(1, \, \hat{\gamma}_2, \, \ldots, \, \hat{\gamma}_n)'$. This is the **only** identified column of B_0^{-1} — the remaining shocks are not recovered (partial identification)
5. **Compute IRFs**: propagate $\hat{b}_1$ through the VAR dynamics and rescale to an economically meaningful shock size (e.g. 25 bp)

Fortunately, we can do all of this the ACB Toolbox with ‘VARopt_exog.ident = ’iv”

A new type of VAR?

- SVAR identification traditionally came from placing restrictions *directly* on B_0^{-1}: short-run, long-run, sign, magnitude constraints, etc.
- Instruments takes a fundamentally different route: Rather than restricting B_0, we use an **observable proxy** z_t to learn about one column of B_0^{-1} indirectly via IV. Since z_t is not the structural shock itself but an observable **proxy**, people call them **proxy SVARs** (see Stock and Watson (2018) and Mertens and Ravn (2013)).
- The result is **partial identification**: one column of B_0^{-1} is recovered; the remaining shocks are left unidentified.

Part 2

Creating Instruments

High-frequency identification in practice

From narrative to high-frequency: a brief history of monetary policy instruments

- **Romer and Romer (1989, 2004)**: Narrative approach — reading FOMC minutes and Greenbook forecasts to date deliberate shifts in monetary policy stance. Widely used but inherently subjective; does not exploit market expectations from financial market prices.

- **Kuttner (2001, JME)**: First to use federal funds futures to decompose changes in the policy rate into *anticipated* and *unanticipated* components. The surprise is the only component that should move interest rates — a clean, market-based measure requiring just one regression on FOMC meeting days.

- **Gürkaynak, Sack and Swanson (2005, IJCB)**: Extended Kuttner to extract two factors — the *target surprise* (current-meeting rate change) and the *path surprise* (changes in expected future rates / forward guidance).

- **Gertler and Karadi (2015)**: Embed the GSS 4-month Fed. Funds Rate futures (FF4) surprise as an external instrument in a proxy SVAR — integrating HFI surprises into the structural VAR framework.

High-frequency identification: the idea

- **Idea**: In a narrow window around a scheduled announcement, the only information that can move asset prices is the announcement itself — there is not enough time for other structural shocks to materialize (unless other announcements or events occur...)
- **Formally**: Let τ denote the announcement time. The instrument is

$$z_t = \Delta p_t^{(\tau-\delta,\, \tau+\delta)}, \qquad \delta \approx 15\text{–}30 \text{ minutes}$$

  where p_t is a futures contract that loads on the shock of interest
- This delivers both IV conditions (almost) by construction:
  1. **Relevance**: the asset responds immediately to the news (*testable via first-stage F-stat*)
  2. **Exogeneity**: within minutes, no macro shock can contaminate the window — but other shocks might...
- Aggregated to monthly (or quarterly) frequency: z_t equals the intra-day surprise in event months, and is coded as missing (NaN) in non-event months.

High-frequency identification: three applications

- **Gertler and Karadi (2015, AEJ:Macro)**: **Monetary policy shock**
  - Instrument: FF4 federal funds futures change in a **30-minute window** around FOMC announcements
  - Identifies genuine policy surprises, free from the price puzzle
- **Känzig (2021, AER)**: **Oil supply news shock**
  - Instrument: Brent crude oil futures change in a **1-hour window** around OPEC announcements
  - Isolates supply-driven oil price movements from demand fluctuations
- **Jarociński and Karadi (2020, AEJ:Macro)**: **Monetary policy vs. information**
  - Instrument: same FF4 futures surprises as in GK (2015), combined with sign restrictions on stock prices
  - Separates *pure policy tightening* (rate $\uparrow$, stocks $\downarrow$) from *Fed information shocks* (rate $\uparrow$, stocks $\uparrow$) — the "Fed information effect"

Also see Choi, Willems, and Yoo (2024) database for MP shock series for SOEs.

Key Idea

Any structural shock transmitted quickly through financial markets can be instrumented using futures price changes around relevant announcements.

Part 3

Gertler and Karadi (2015)

Monetary policy through the credit channel

Gertler and Karadi (2015, AEJ:Macro): Overview

Question: What are the effects of a contractionary monetary policy shock on credit costs and economic activity?

Motivation: Recursive SVARs generate the **price puzzle** for monetary policy — a timing restriction on who sees what at impact is not credible when policy responds to many variables simultaneously.

Contribution: Use high-frequency movements in *federal funds futures* around FOMC announcements as an external instrument. The instrument isolates genuine policy surprises — information released in a 30-minute window around FOMC announcements that is unlikely to reflect aggregate demand news.

Finding: Monetary tightening raises short rates and the excess bond premium (a measure of credit conditions), while compressing output and prices — consistent with a credit channel of transmission.

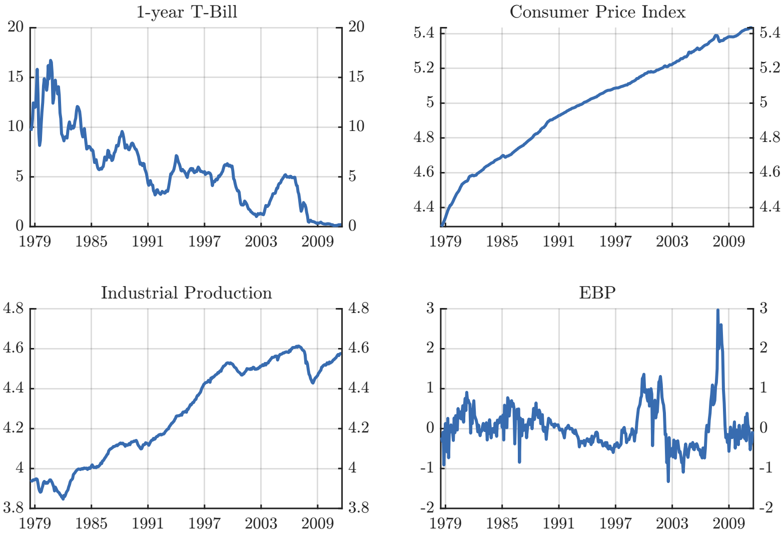
Gertler and Karadi (2015): The VAR and instrument

- **Monthly SVAR**, 1979:M7–2012:M6; United States. $p = 12$ lags, constant.
- **Four variables** (ordered first to last):
  1. gs1_t = 1-year Treasury bill rate (%) $\leftarrow$ **instrumented variable**
  2. lcpi_t = log CPI $\times$ 100
  3. lip_t = log industrial production $\times$ 100
  4. ebp_t = excess bond premium (%)
- **Instrument**: **FF4** futures price change in a 30-minute window around each FOMC announcement:

  $$z_t = -\Delta p_t^{(\tau-10,\, \tau+20)}$$

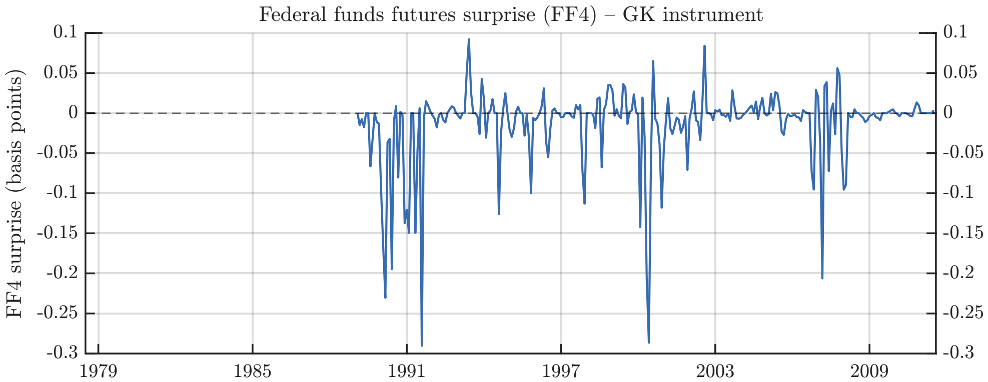
  where p_t is the FF4 futures price and τ is the announcement time. The negative sign converts from price to rate ($\uparrow$ futures price $=\downarrow$ expected funds rate). FF4 captures surprises about the *near-term path* of rates, not just the meeting-day decision. Non-FOMC months coded as missing (NaN).
- The 1-year rate is ordered **first** so that the proxy SVAR instruments the first equation's residuals.

Gertler and Karadi (2015): Data



Ex: When is monetary policy most contractionary? How does the EBP evolve during the 2008 financial crisis?

Gertler and Karadi (2015): Instrument and first-stage



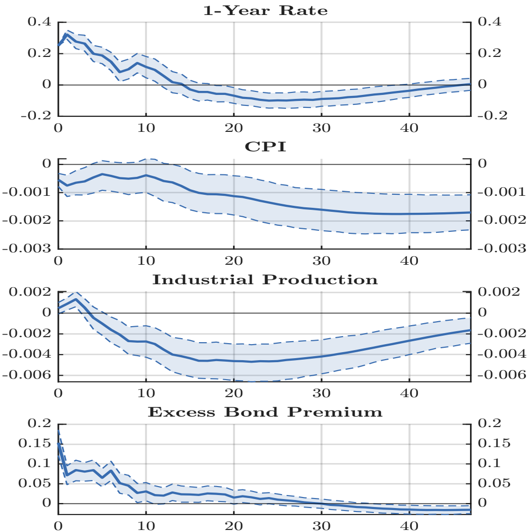
First-stage: GS1 residual regressed on FF4 surprise (FOMC months only)

Coefficient	F-statistic	R^2	N
1.1341	21.52	0.0743	270

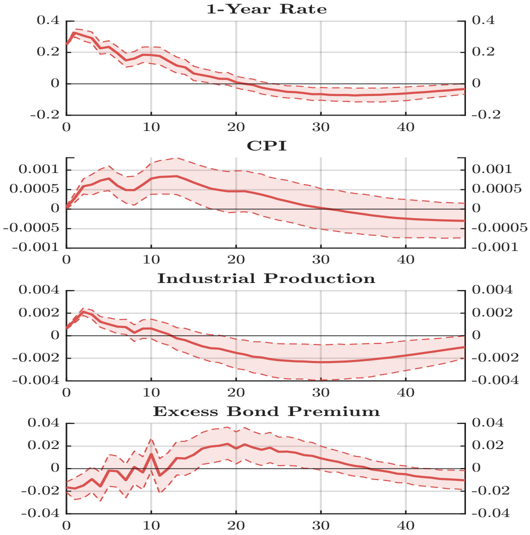
Ex.: Is the Stock–Yogo (2002) rule of thumb satisfied?

Gertler and Karadi (2015): IRFs — Proxy SVAR vs Cholesky

Proxy SVAR



Recursive VAR



Part 4

A Simpler Approach

Just use a VARX...

A simpler estimator

- Plagborg-Møller and Wolf (2021) and Paul (2020) note that instead of using z_t as an *instrument* for the residuals, we can include it **directly** as a contemporaneous exogenous regressor in every VAR equation (**VARX**):

$$x_t = c + \sum_{j=1}^{p} \Phi_j x_{t-j} + \delta z_t + e_t$$

- The OLS estimate $\hat{\delta}$, scaled to the same size shock, gives the first column of B_0^{-1} — much simpler!

- Plagborg-Møller and Wolf (2021) prove that VARX and proxy SVAR give asymptotically identical IRFs and conditions for agreement in finite samples, and Paul (2020) uses this idea in the TVP-VAR setting; see Terrell, Haque, Cross and Doko Tchatoka (2026) for an application to monetary policy shocks and exchange rate dynamics in small open economies.

Plagborg-Møller and Wolf (2021): The equivalence result

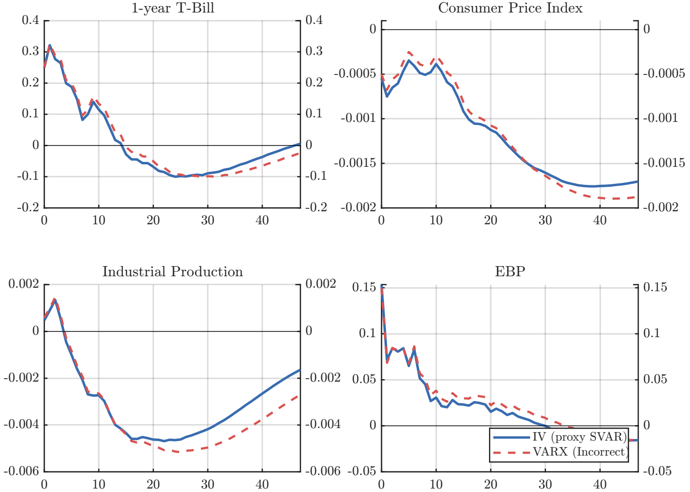
In finite samples, the two estimators agree when:

1. **Common sample**: Both use the same set of observations
2. **Pre-orthogonalization**: The instrument is orthogonal to the VAR lag matrix X:

$$z^{\perp} = M_X z, \qquad M_X = I - X(X'X)^{-1}X'$$

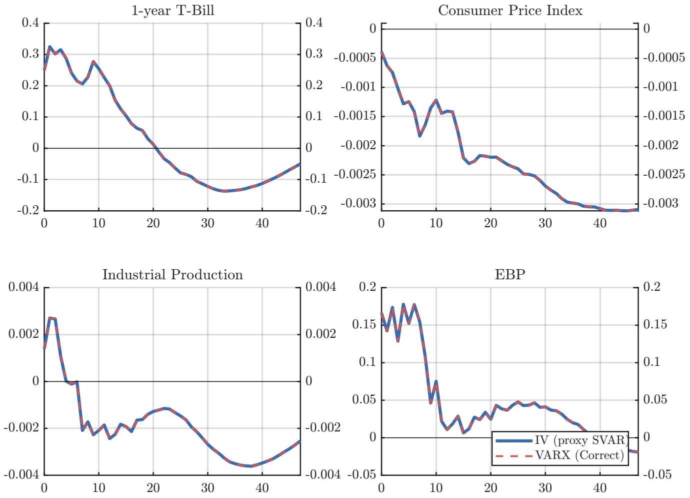
After doing this, implement in ACB Toolbox with ‘VARopt_exog.ident = ’exog”

VARX (incorrect): full sample, zero-filled instrument



If IV (solid) drops non-FOMC months OR VARX (dashed) zero-fills them then the PMW equivalence conditions are violated and produce visibly different IRFs.

VARX (correct): FOMC subsample, orthogonalized instrument



Restricting both estimators to the FOMC subsample and pre-orthogonalizing the instrument to the lag matrix, the two IRFs coincide.

External instruments: Pros and Cons

Pros

- Theory-grounded: high-frequency surprises are a credible, institutional argument for exogeneity
- Avoids arbitrary recursive ordering — no timing assumptions about who sees what
- Point-identified: unique IRFs (unlike sign restrictions)
- Well-suited to combining with narrative evidence on instrument quality

Cons

- Instrument exogeneity is *never* directly testable
- Weak instruments (low F) produce large finite-sample bias
- Instruments available only for a subset of shocks and periods
- Proxy SVAR drops non-announcement months, shrinking the effective sample

Summary: External Instruments

1. **Simultaneity is the fundamental problem**: OLS on observational data identifies neither supply nor demand. An instrument that shifts only one curve restores identification — the same logic applies to SVARs.

2. **Two conditions**: Relevance ($F > 10$, testable) and exogeneity (untestable, requires institutional argument). High-frequency identification satisfies both via market microstructure.

3. **High-frequency identification**: Asset price changes in a narrow window around scheduled announcements (FOMC, OPEC, etc.) deliver credible instruments. Kuttner (2001) pioneered this for monetary policy; Gürkaynak et al. (2005) extracted the FF4 path factor; GK (2015) embedded it in a proxy SVAR.

4. **PMW equivalence**: The VARX (exogenous variable) formulation is asymptotically equivalent to the proxy SVAR under the same identifying assumptions. Much easier to implement when not using the toolbox...

Karagedikli and Kuttner (2025): Do funds rate surprises matter anymore?

Question: As the Fed has become more transparent, do FF4 futures surprises still constitute a valid instrument for monetary policy shocks?

Key findings: Except during episodes of drastic rate cuts, markets have rarely been surprised by FOMC decisions in recent years — greater transparency has compressed the meeting-day surprise. Fed chair speeches and macro data now drive most of the variation in rate expectations. Communication has had mixed success: minutes and testimony sometimes move expectations in the *wrong* direction.

Implications for proxy SVARs:

- Near-zero surprise series $\rightarrow$ weak first-stage F-statistic $\rightarrow$ no proxy SVAR...
- The remaining surprise variation may mix genuine policy shocks with communication noise — threatening exogeneity
- What is the right instrument for the post-forward-guidance era?

New ideas are needed here!

Thank you!